

PRISM

Predictive Risk Insights & Scoring Model

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1. Overview

This document specifies the features of PRISM — a credit scoring and risk assessment prototype developed under the TIH IoT CHANAKYA Fellowship 2025. It is the reference for development, milestone reviews, and TRL-4 evaluation.

PRISM targets India's 190 million credit-invisible citizens by combining traditional bureau data with behavioural signals, NLP, explainable AI, fraud detection, and document verification in a single ML-driven pipeline.

2. Feature specification

Priority: Must = required for TRL-4; Should = targeted within timeline; Could = stretch goal.

Feature ID	Feature	What it does	Key inputs	Key output	Priority
Core Scoring					
CS-01	Credit score	Composite risk score (300–900) from multi-modal data	Bureau summary, EPFO, utility records, bank statement	Score + risk tier (Low/Med/High/Very High)	Must
CS-02	Confidence flag	Flag low-certainty predictions for human review	CS-01 prediction probabilities	Confidence % — routed to manual queue if <60%	Must
Behavioural Analysis					
BA-01	Cashflow stability	Measure income regularity from bank statement	12-month credit/debit entries	Stability index 0–1; trend label	Must
BA-02	Utility discipline	Score utility payment regularity as credit proxy	Electricity, water, telecom payment records	Discipline score 0–100; late payment count	Must
BA-03	Employment continuity	Verify employment stability via EPFO records	EPFO contribution months, gaps	Employment score; gap flag (>3 months)	Must
NLP Module					
NL-01	Loan purpose class.	Classify stated loan purpose into risk categories	Free-text purpose field	Category: Productive / Consumption / Speculative	Should
NL-02	Sentiment scoring	Score borrower's sector using financial news NLP	Sector news feed, borrower's industry	Sentiment score –1 to +1; risk modifier	Could
Explainability					
EX-01	SHAP reason codes	Explain each score with top contributing features	Trained XGBoost model + feature vector	Top-5 positive and negative reason codes	Must
EX-02	What-if analysis	Simulate score change when inputs are modified	Current feature vector + hypothetical overrides	Revised score + delta from current	Should
Fraud Detection					

FD-01	Loan stacking	Detect multiple applications across lenders	PAN, application timestamps	Stacking flag; count in 30-day window	Must
FD-02	Income anomaly	Flag mismatch between stated and actual income	Stated income vs bank statement average	Overstatement ratio; severity flag	Must
FD-03	Identity check	Detect implausible identity-data combinations	Age, PAN year, EPFO start, credit history age	Plausibility score 0–100; flag if <40	Should
Document Verification					
DV-01	OCR validation	Extract and validate fields from uploaded documents	PDF/image bank statement or salary slip	Extracted fields + validation flag per field	Must
DV-02	Metadata check	Detect post-creation document tampering	File creation date, modification date, author	Tamper flag; anomaly type	Should
Compliance					
CL-01	Consent capture	Record borrower consent with audit trail	Consent type, borrower ID, timestamp	Consent record; audit log entry	Must
CL-02	PII masking	Mask personal identifiers in all outputs and logs	Raw PAN, Aadhaar, account number, phone	Masked output; PII-free log entries	Must
System					
SL-01	REST API	Expose all PRISM features via REST endpoints	JSON borrower payload + auth token	JSON response: score, flags, reason codes	Must
SL-02	Lender dashboard	Web UI for credit officers to view borrower profiles	API response data	Score gauge, SHAP chart, fraud flag cards	Should

3. Non-functional requirements

NFR ID	Category	Requirement
NFR-01	Performance	Score generated in ≤ 3 seconds for 95% of requests on sample dataset
NFR-02	Fairness	Default rate difference across gender and geography subgroups $\leq 5\%$ on synthetic test set
NFR-03	Explainability	Every CS-01 prediction must return ≥ 3 SHAP reason codes in API response
NFR-04	Auditability	Every score logged with borrower ID, timestamp, and model version; retained 90 days
NFR-05	Security	All PII encrypted at rest (AES-256); all API communication over TLS 1.3
NFR-06	Compliance	Consent recorded before any data access; PII masked in all outputs; aligned with DPDPA 2023

NFR-07	Reliability	Failure of any single module must not crash the pipeline; fallback to available modules with degraded confidence
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